

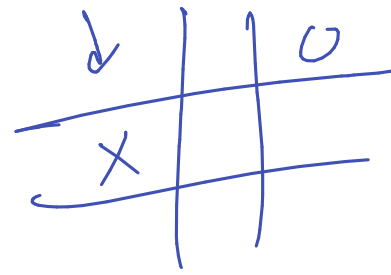
Classroom code: fjvbnj6

Reinforcement Learning

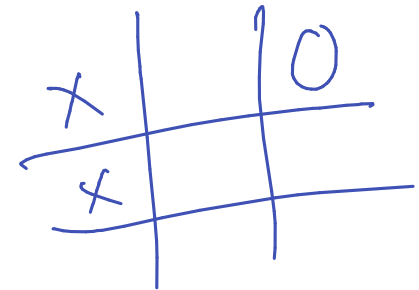
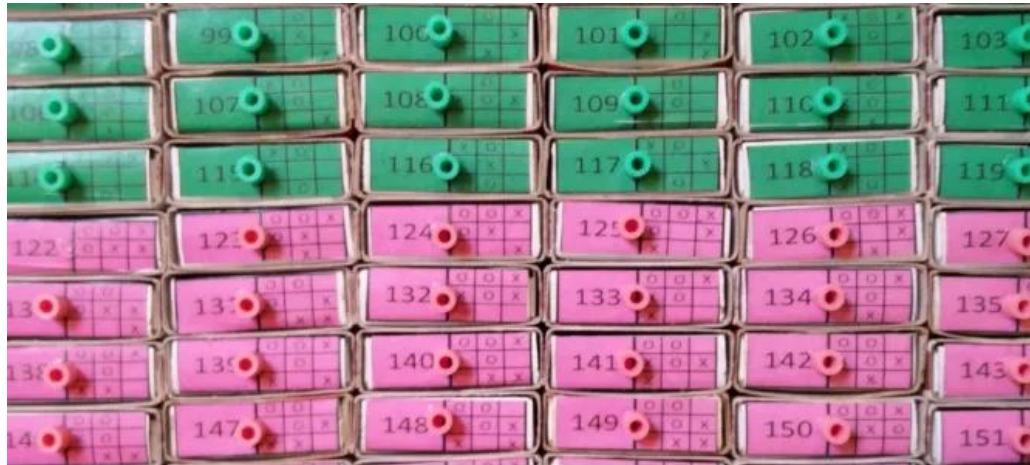
Approximate Dynamic Programming/ Neuro Dynamic Programming

Sanjit K. Kaul

Mathematical Games



- 1961: Matchbox Educable Naughts and Crosses Engine (MENACE)
- Teaching a machine to play Naughts and Crosses (Tic-Tac-Toe)
 - Donald Michie, Biologist at University of Edinburgh, used 304 matchboxes
- One matchbox per tic tac toe position



Source: <http://chalkdustmagazine.com/features/menace-machine-educable-noughts-crosses-engine/>

Mathematical Games

- One matchbox per tic tac toe position [***State***]
- Each matchbox contains multiple beads of different colors
 - Each color corresponding to a possible move [***Actions***]
 - Shake the matchbox to randomly draw a bead
- More details
 - <http://cs.williams.edu/~freund/cs136-073/GardnerHexapawn.pdf>

Mathematical Games

- Nudge the machine to make winning moves
 - Remove (reduce in number) beads that led to a defeat in a game
 - Increase beads that led to a victory/ draw
- More details
 - <http://cs.williams.edu/~freund/cs136-073/GardnerHexapawn.pdf>

The Optimization Problem

- Stochastic and Multi-stage (multi-time-step)

- **Not** one shot optimization

$$\underset{x}{\text{Minimize}} \quad f(x)$$

- **One shot [Have you come across this before?]**

- A priori deciding that you will drive in the right most lane
- A priori deciding the sequence of lane changes you will stick to

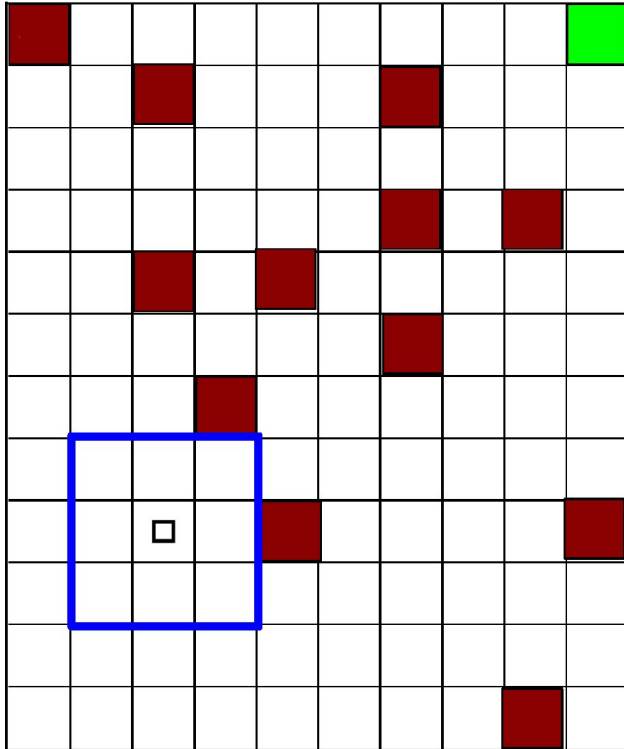
The Optimization Problem

- Stochastic and Multi-stage (multi-time-step)
- **Not** one shot optimization
 - **One shot [Have you come across this before?]**
 - A priori deciding that you will drive in the right most lane
 - A priori deciding the sequence of lane changes you will stick to
 - **Multi-Stage**
 - Change lane while you drive based on your perceived lane occupancy
 - Beneficial as the lane conditions are stochastic (not deterministic)
 - Multi-stage allows adapting to the current conditions
 - *The current matchbox*

Where Does Such Decision Making Appear?

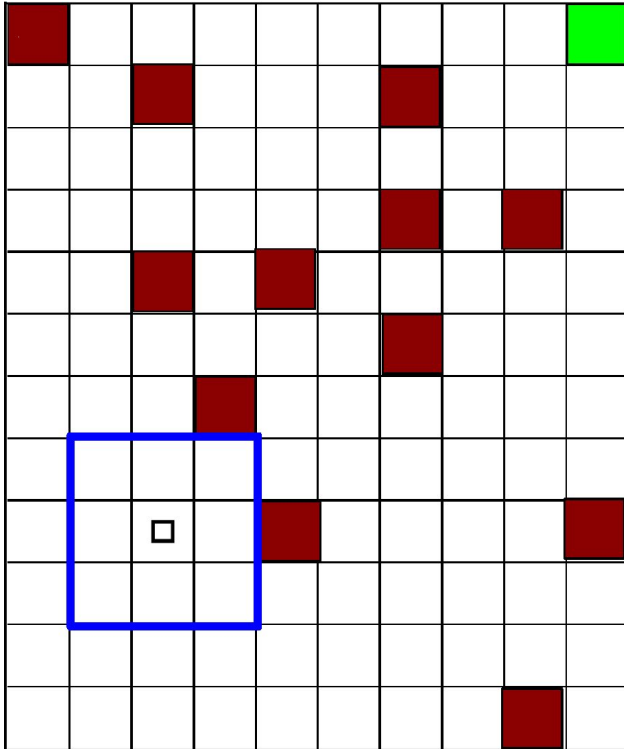
- Operations Research
 - Inventory control
 - What's stochastic here?
- Artificial Intelligence
 - Play games like humans, robots learn to navigate their environment
- Optimal Control
 - Path planning
 - Network optimization

The Grid World



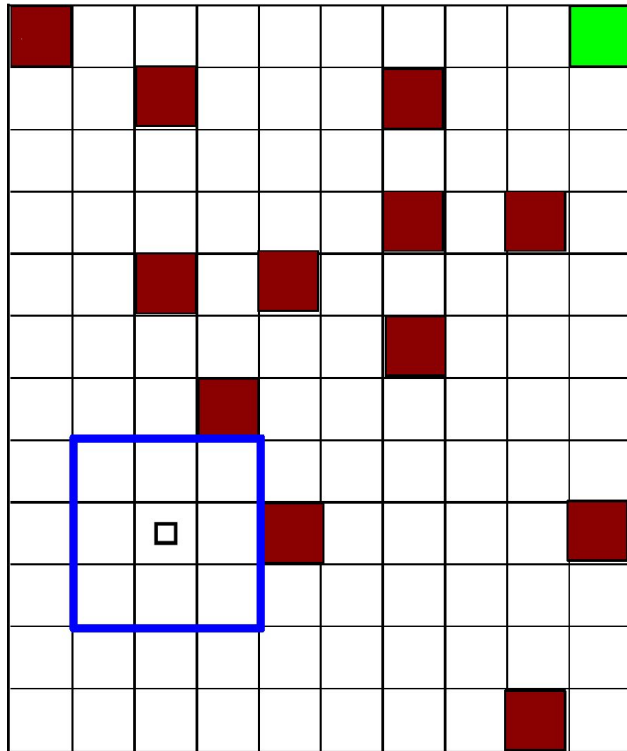
- Environment
- Discrete Time
- Agent
- State/ Observation
- Actions
- Reward

The Grid World



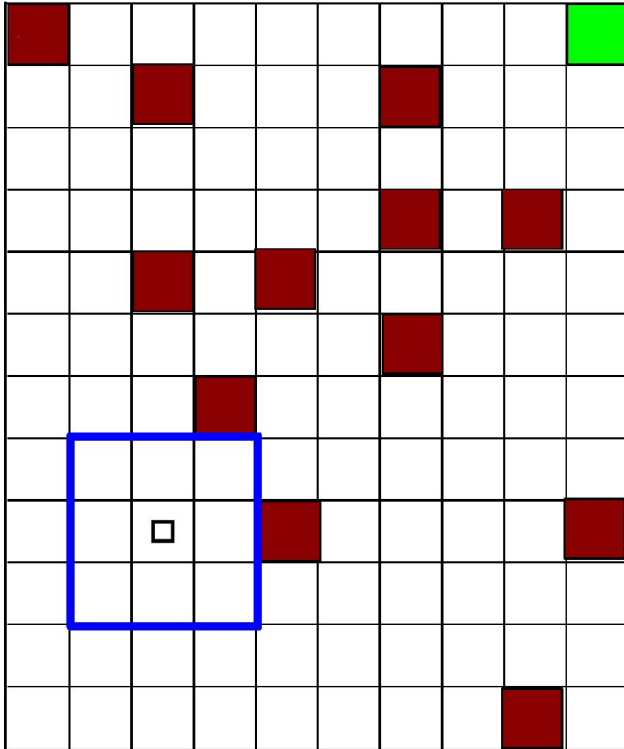
- State is a perfect summary of the current environment
 - All that the agent needs at a given time to choose the best possible action.
 - No additional information can help, given the state.
- State is Markov

The Grid World



- Example State

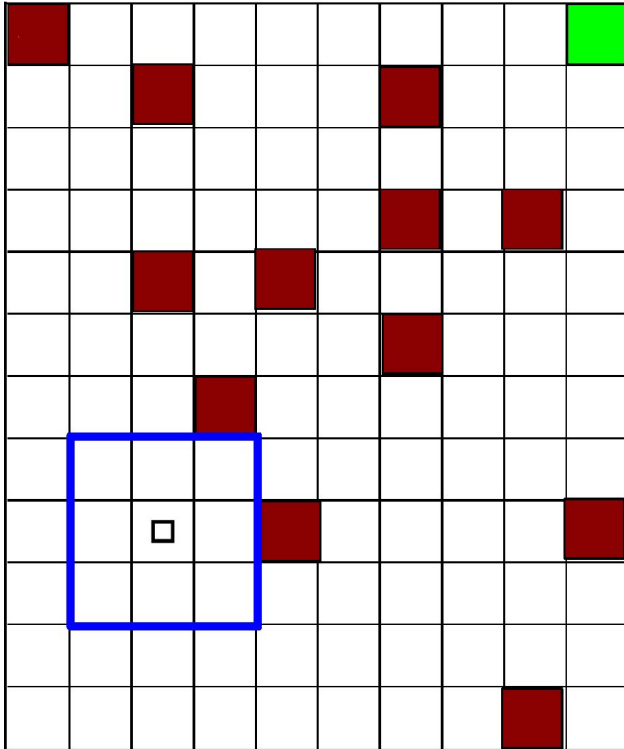
The Grid World



- State is a perfect summary of the current environment
 - State is Markov

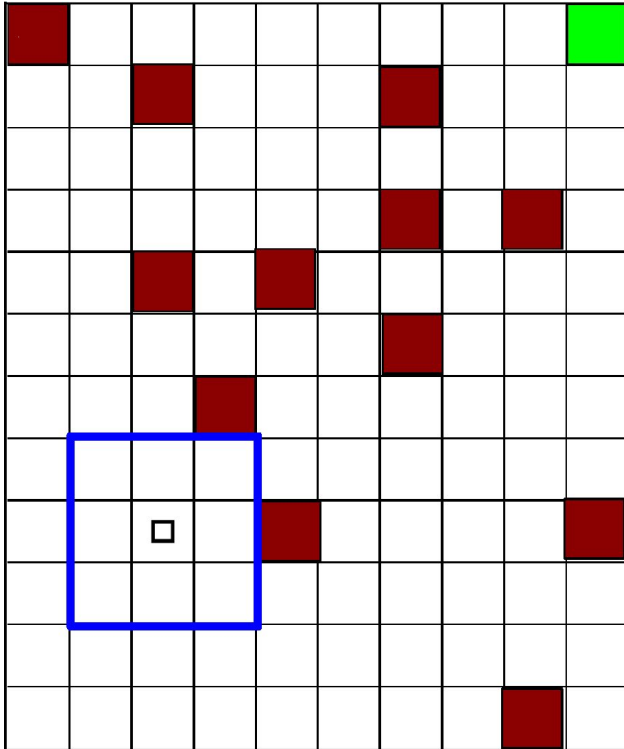
- Often, an agent can't see the state
 - Has access to observations
 - Imperfect state information

On the State



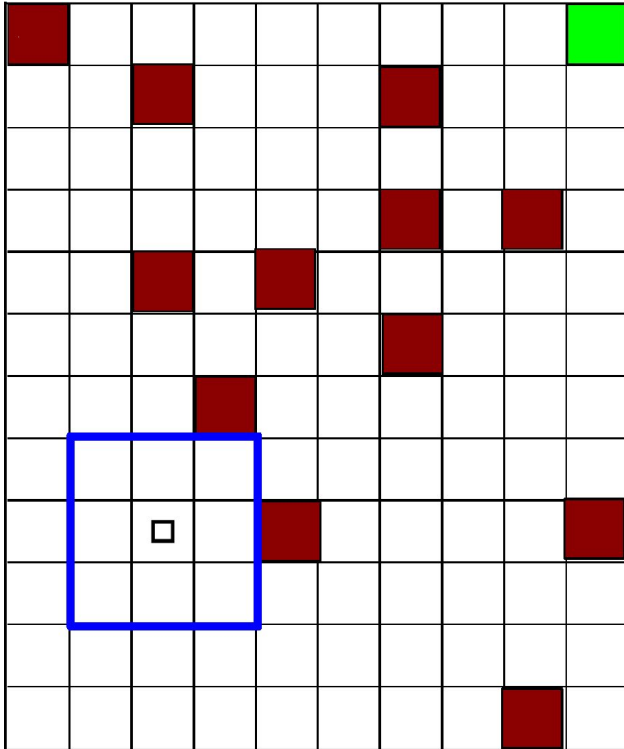
- What if the grid is just an instance drawn in an *iid* manner?
- What is a good choice of state?

On the State



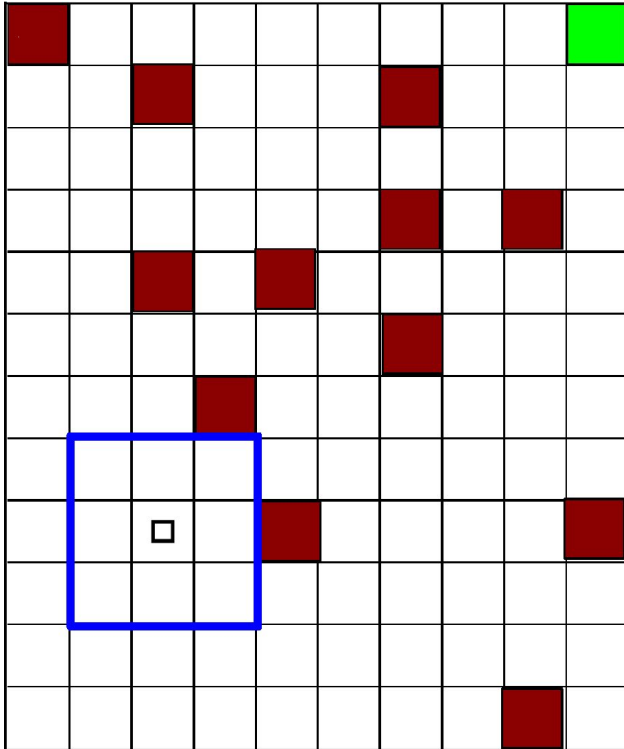
- What if the grid is just an instance drawn in an *iid* manner?
- What is a good choice of state?
 - Agent's location on the grid?
 - **Entire grid occupancy (obstacle/ free) and the agent's location?**
 - Only the occupancy of the cell where the agent is currently?
 - Location of the agent and occupancy of all cells within the ~~purple~~ *blue* rectangle?

The Grid World



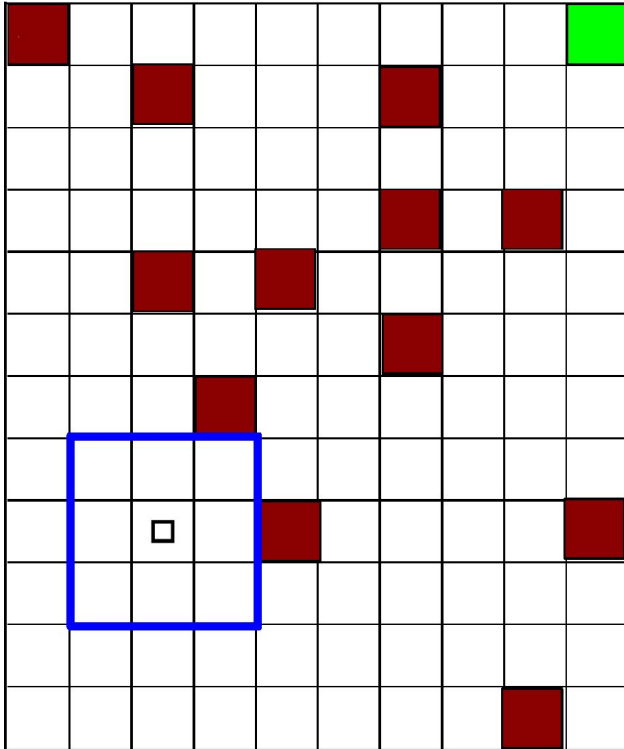
- Agent ***observes*** state and chooses its action

The Grid World



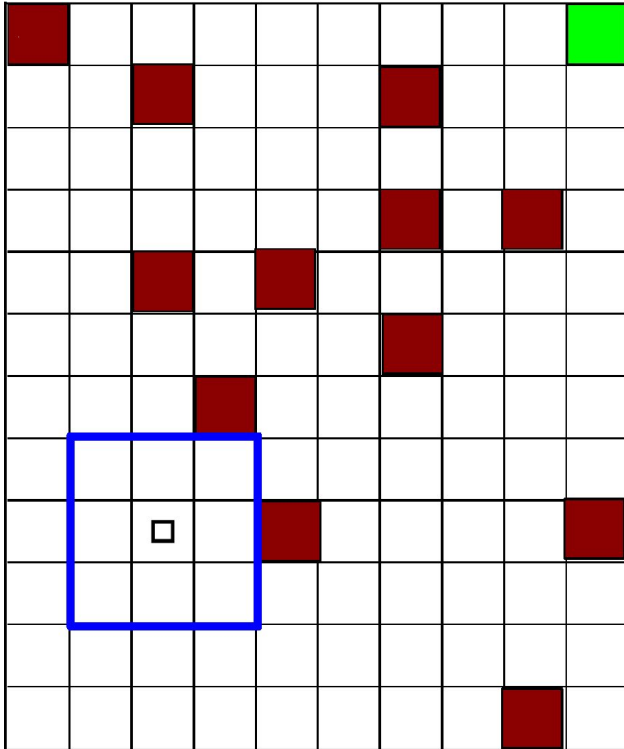
- Agent uses a policy function to:
 - The functional mapping between states and actions
- Agent would like to *learn* the *optimal* policy function

The Grid World



- Rewards from the environment
 - Agent receives a reward after every action

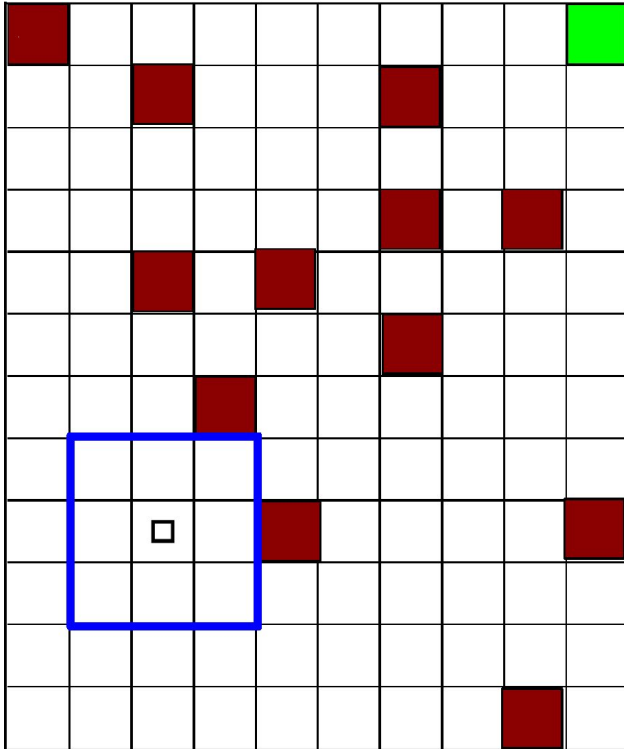
The Grid World



- Rewards from the environment
 - Agent receives a reward after every action

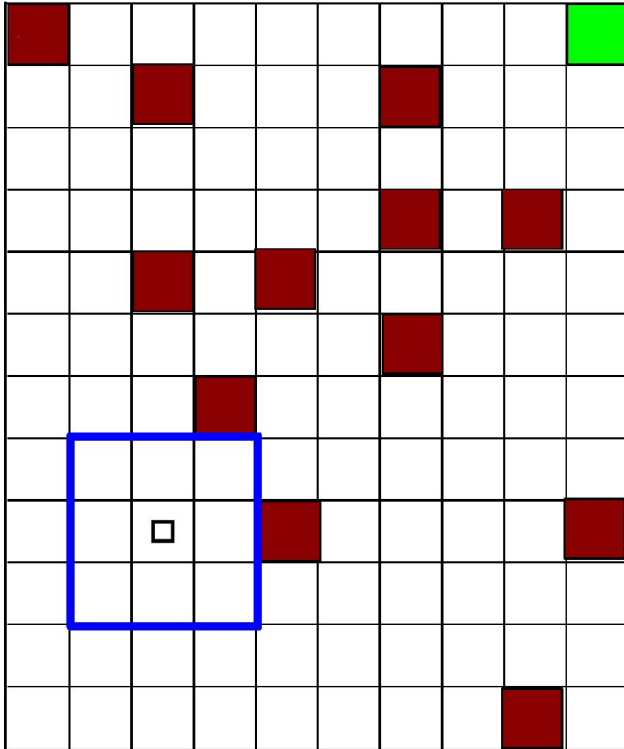
- Rewards help the agent to improve its policy

The Grid World



- You (the human/RL practitioner) may need to design rewards
- Nudge the agent to choose beneficial actions...
 - -1 reward for moving into an occupied cell
 - 0 for moving into a free cell
 - 1 for moving into the goal!

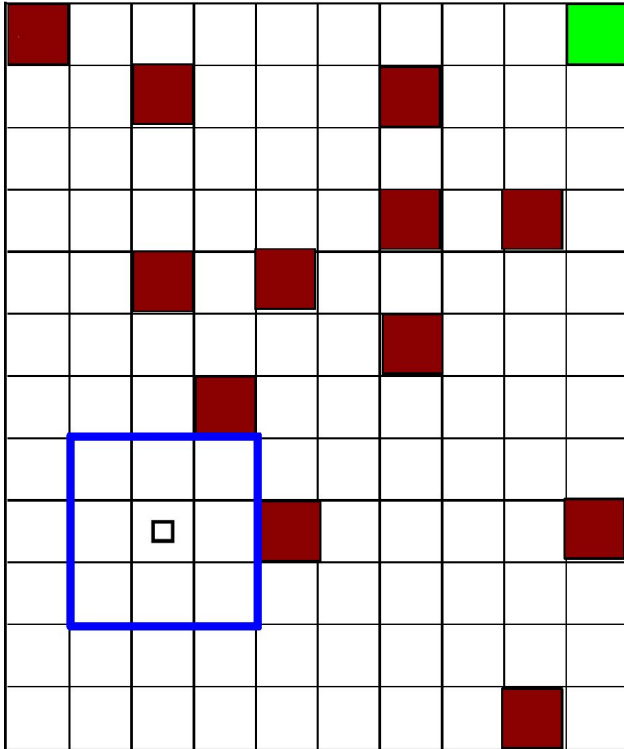
The Grid World



Uncertainty/ Randomness/ Stochastic Environment

- Given the current state and an action:
 - The received reward and next state is in general random (governed by a PMF/PDF)

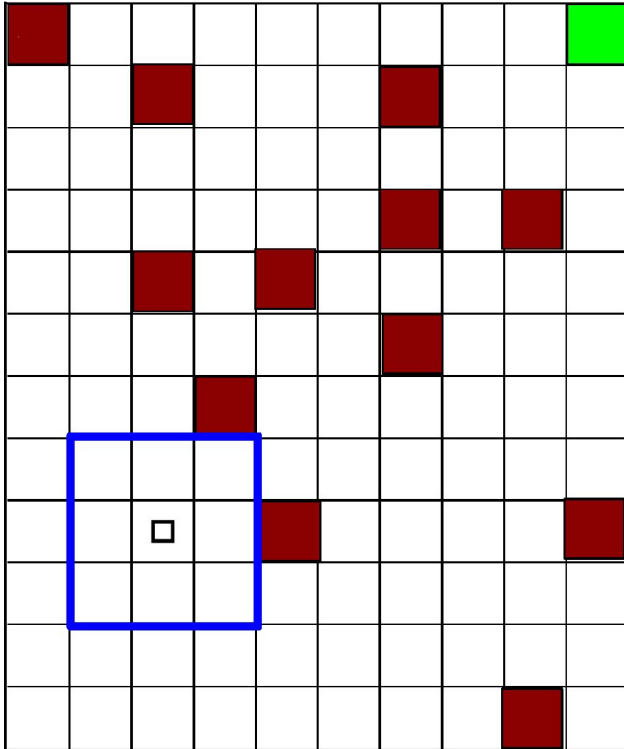
The Grid World



Uncertainty/ Randomness/ Stochastic Environment

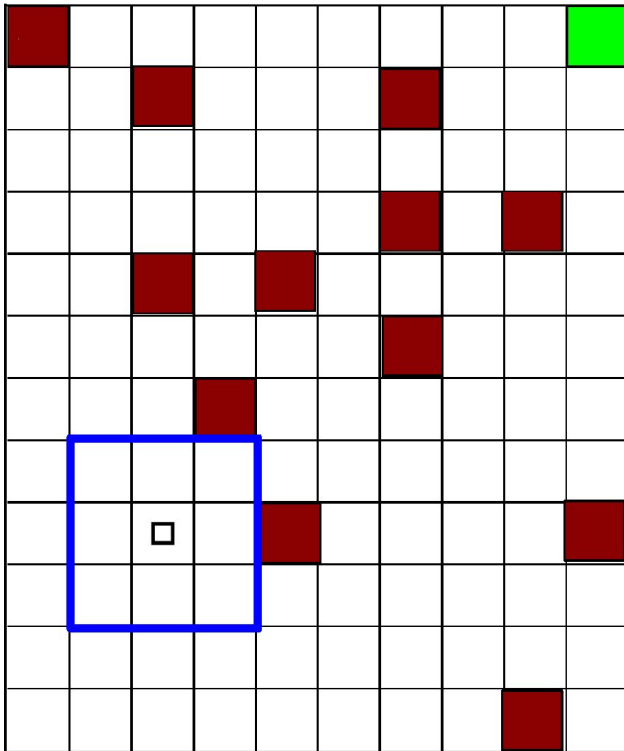
- Even the policy may be a distribution over actions!

The Grid World



- Tasks may be finite time steps long, episodic, infinite

The Grid World



- **Episodic**: Task is guaranteed to end in a finite but random number of steps
- Random **Return** obtained at the end of the episode, starting at time t

$$G_t = R_{t+1} + R_{t+2} + \dots + R_T$$

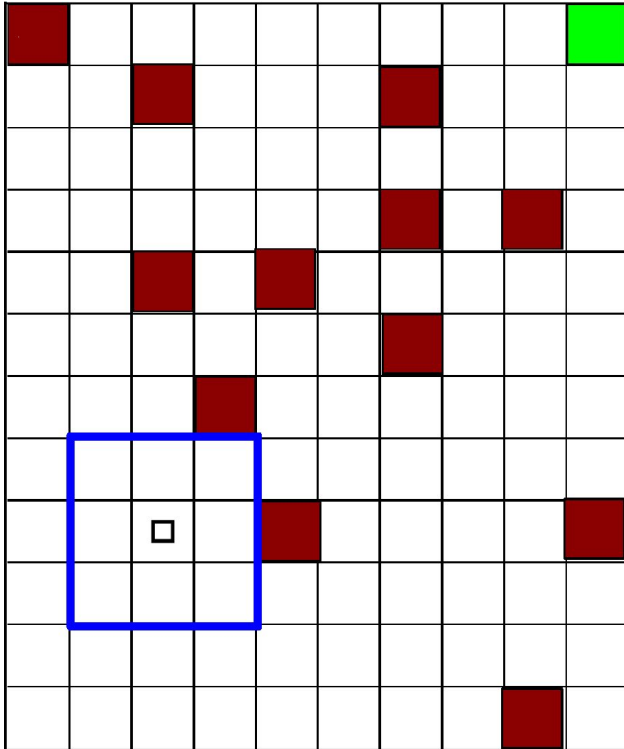
$$\text{Maximize } E[G_0 | S_0 = s_0]$$

$$[\pi_0, \pi_1, \dots]$$

$$\text{Maximize } E[G_0 | S_0 = s_0]$$

$$\pi$$

The Grid World



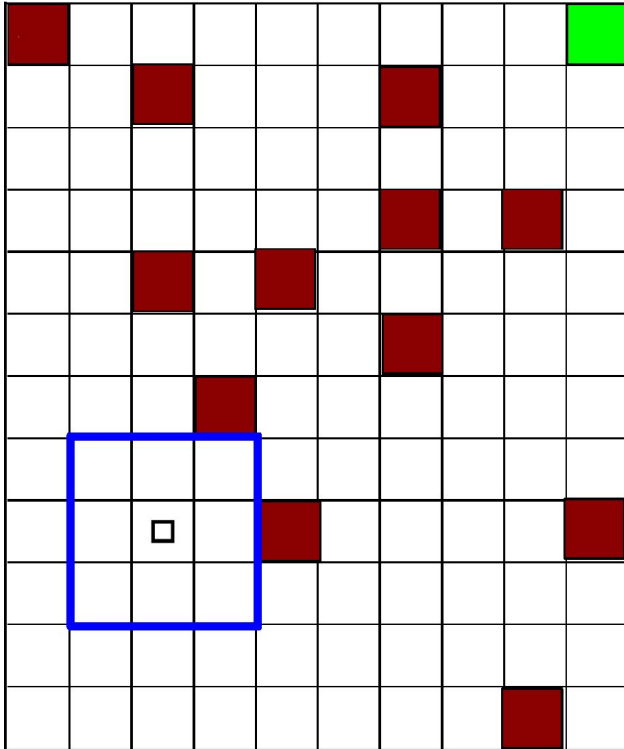
- **Infinite**: Task continues forever... for an infinite number of time steps

- Random **Return** obtained at the end of the episode, starting at time t

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \\ + \gamma^{k-1} R_k + \dots$$

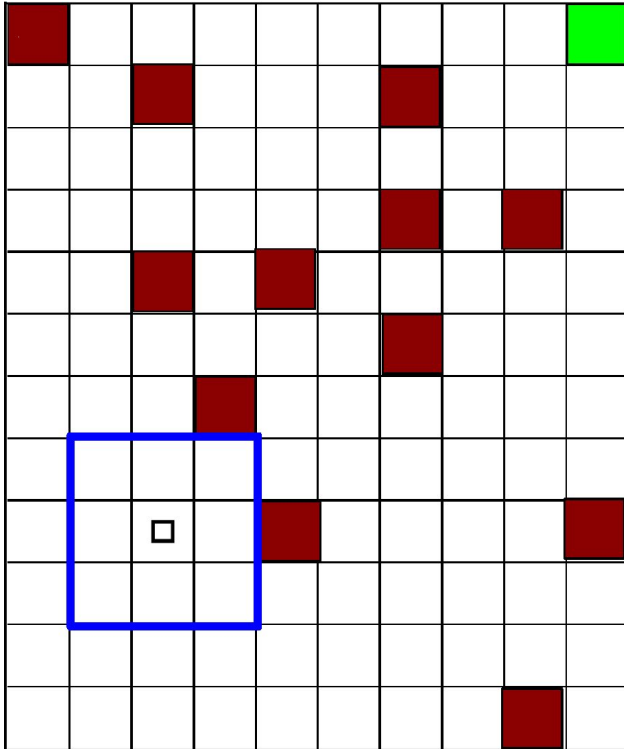
- G_t in terms of G_{t+1} ?

The Grid World



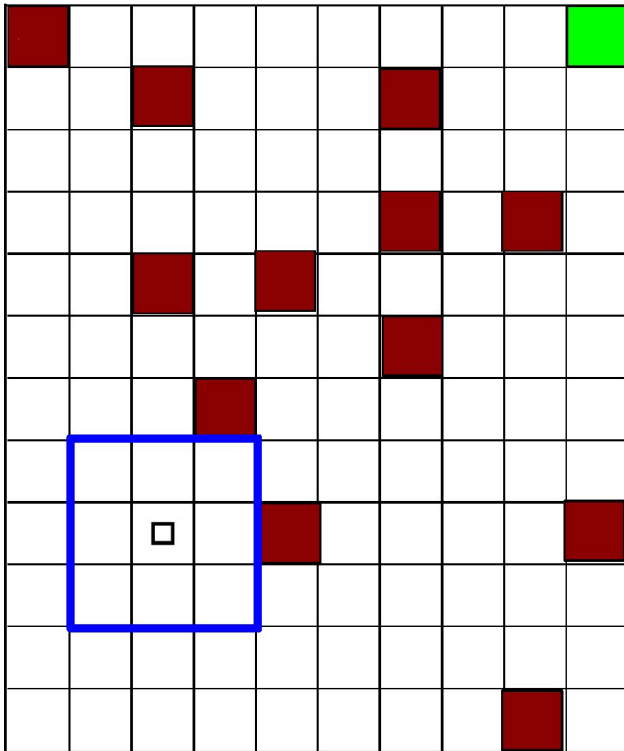
- What kind of a task is the Grid World?

The Grid World



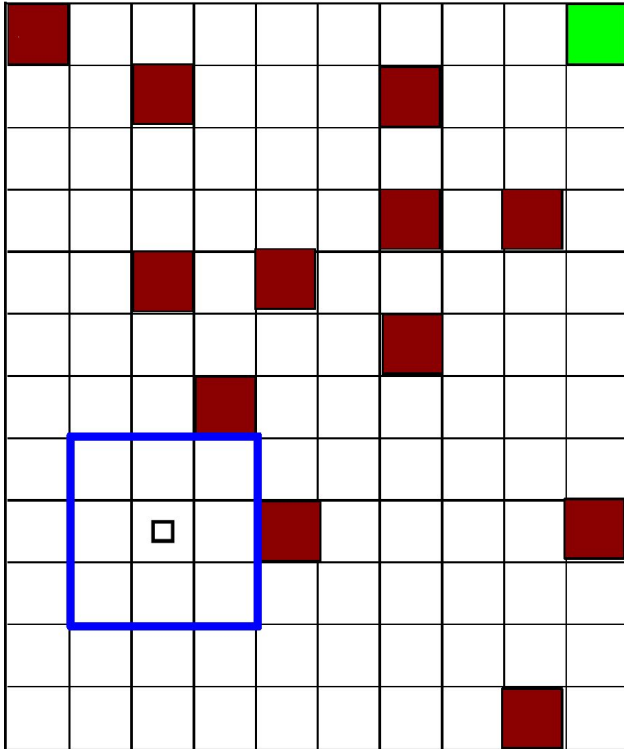
- Consider the agent navigating the grid world
- Over time we get:
 - sequences of random states, actions, and rewards
- Example sequences?

The Grid World



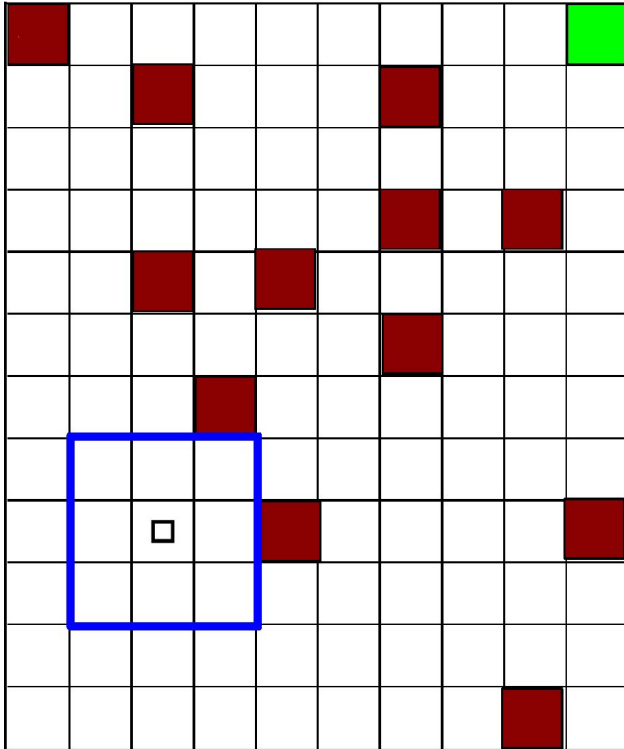
- Agent obtains a sequence of random rewards
- Little sense in optimizing random variables
- Instead optimize *expected* values of random variables
- **Agent's goal:** Maximize *expected* sum reward (expected **Return**) starting in any state

The Grid World



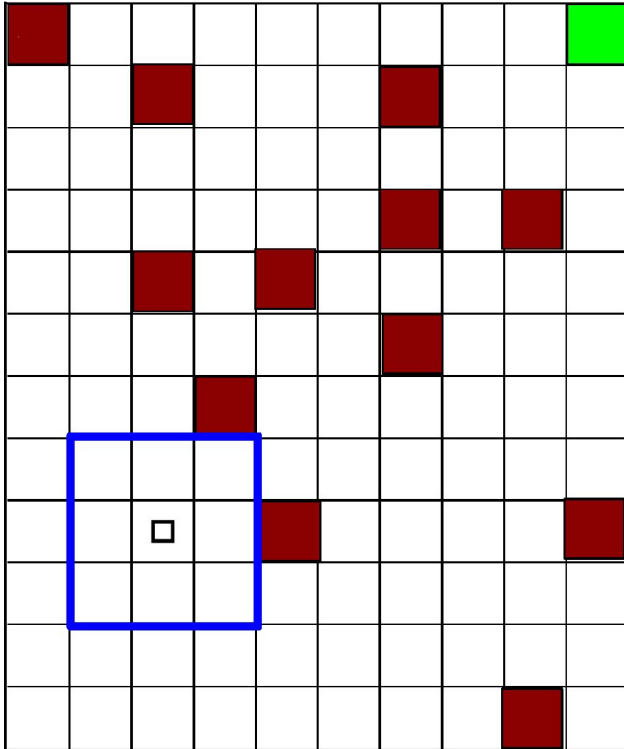
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The Grid World



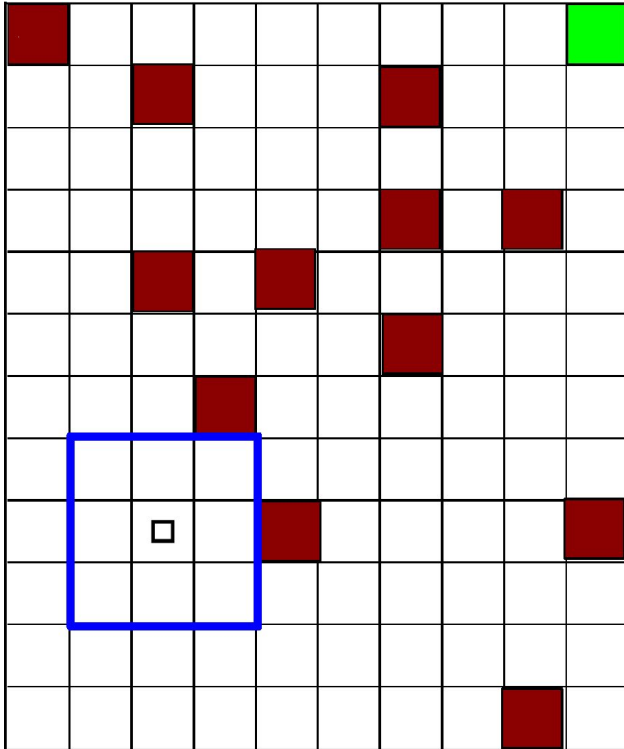
- **Agent's goal:** Maximize *expected Return* starting in any state

The Grid World



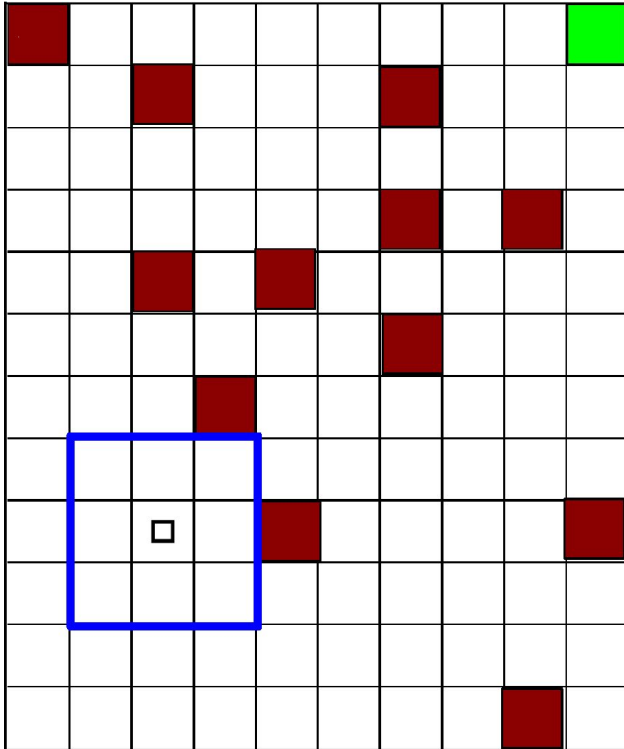
- RL algorithms help the agent learn an optimal policy
 - At every time step agent gives the observed state to the policy
 - Policy tells the agent the action it must perform

The Grid World



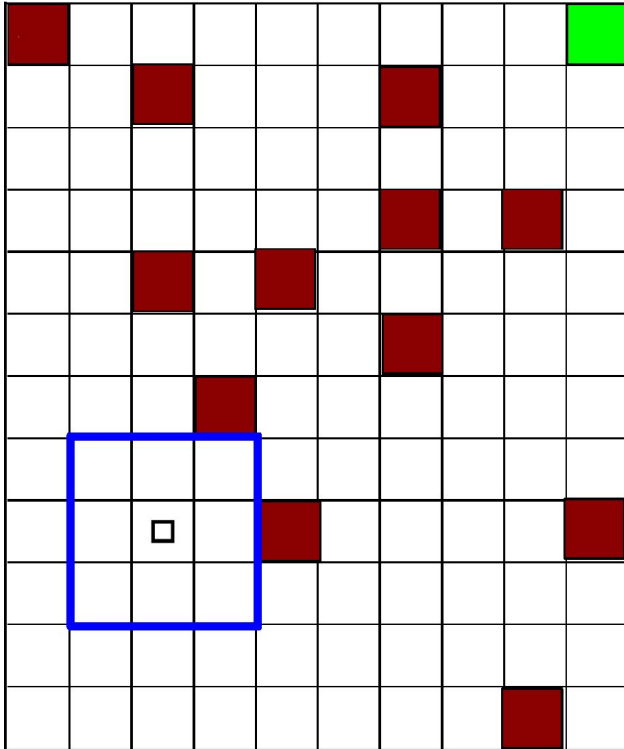
- The optimization problem:
- An optimal policy maximizes the expected sum reward

The Grid World



- How would one-shot optimization look like?

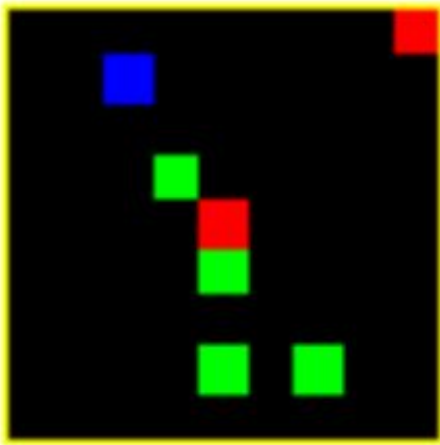
The Grid World



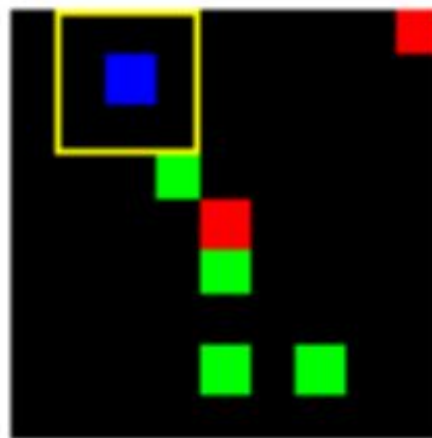
- How would one-shot optimization look like?
- The agent sticks to the resulting *optimal* sequence of actions
 - Unable to adapt its actions to what it sees when it interacts with the environment

The Grid World - Learned policies

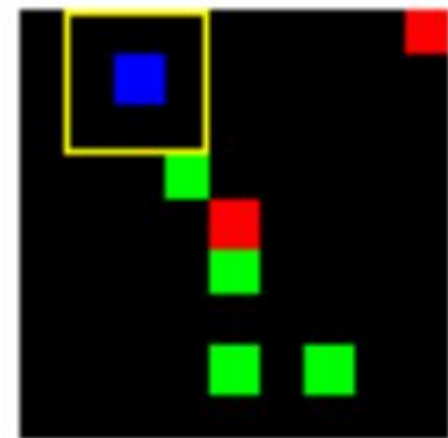
Perfect Markov
State



Imperfect
State
(Observation)



History of
Observations



Prerequisites

- Probability

- Undergraduate level should suffice
- Discrete RV(s), PMF, Expectation, Conditional Expectation and etc.

- Programming

- Familiarity with Python ~~will help~~ *is mandatory*

Textbook

- Reinforcement Learning: An introduction by Sutton and Barto
 - Available online [Second Edition ISBN: 978-0-262-19398-6]

Grading (Tentative)

- ~~In Classroom Q&A: 5%~~
 - Will count only if you attempt 75% of it
 - Different lectures may have different amounts of Q&A
 - Timely submission over Google Classroom for your answer to count
 - Typically multiple choice or paragraph answers
- All Exams (About four): 75% ($12.5 \times 2 + 25 \times 2$) — 80%
- May include a coding exam
- Projects + Assignments: 20 % [Max weight to off-line / team work]

2023:
 $15\%, 15\%, 25\%, 25\%$
 Quizzes Mid & End
 20% Project.

Course Coverage

- Multi-armed bandits
- Dynamic programming algorithm
- Tabular Methods
- Approximation Methods
- Policy Gradient
- Deep RL (via papers)

Probability

Review

Some slides courtesy Roy Yates and David Goodman
as a part of their book's related materials

The Basics

- Sample space
 - Set of all possible outcomes
- Events
- Event space
- Mutually exclusive and collectively exhaustive events
- Bayes' Theorem and Law of total probability
- Independent events

Discrete Random Variables

- PMF
 - Definition
 - Conditioning on an event
- Expectation
 - Definition and properties
 - Conditioning on an event
- Expectation of a function of a RV

Quiz 2.9

On the Internet, data is transmitted in packets. In a simple model for World Wide Web traffic, the number of packets N needed to transmit a Web page depends on whether the page has graphic images. If the page has images (event I), then N is uniformly distributed between 1 and 50 packets. If the page is just text (event T), then N is uniform between 1 and 5 packets. Assuming a page has images with probability $1/4$, find the

- (1) conditional PMF $P_{N|I}(n)$
- (2) conditional PMF $P_{N|T}(n)$
- (3) PMF $P_N(n)$
- (4) conditional PMF $P_{N|N \leq 10}(n)$
- (5) conditional expected value $E[N|N \leq 10]$
- (6) conditional variance $\text{Var}[N|N \leq 10]$

Definition 7.1 Sample Mean

For iid random variables $X_1, \dots, X_n$ with PDF $f_X(x)$, the sample mean of X is the random variable

$$M_n(X) = \frac{X_1 + \dots + X_n}{n}.$$

Theorem 7.8 Weak Law of Large Numbers

If X has finite variance, then for any constant $c > 0$,

(a) $\lim_{n \rightarrow \infty} P[|M_n(X) - \mu_X| \geq c] = 0,$

(b) $\lim_{n \rightarrow \infty} P[|M_n(X) - \mu_X| < c] = 1.$

Theorem 7.3 Chebyshev Inequality

For an arbitrary random variable Y and constant $c > 0$,

$$P [|Y - \mu_Y| \geq c] \leq \frac{\text{Var}[Y]}{c^2}.$$

Why Do We Care?

- Think a sum of a very large number of numbers
- Now think Expectations

Stationary (Environment)

- Stationarity in the probabilistic sense
 - Stationary doesn't imply that nothing changes
 - It means that the *probability distributions of interest* aren't a function of absolute time
- Mathematically

Solving Linear Equations

- Think minimizing squared error
- Inverse based and Iterative solutions

Gradient Descent

Multi Armed Bandits

Feedback From the Environment to The Agent

- Instructive Feedback
- Evaluative Feedback

The k -Armed Bandit

- k arms
- Selecting an arm gives a *reward*
- Finite number of attempts
- Maximize expected sum reward

The k -Armed Bandit

- Suppose there are 10 arms
- Arms 1 - 9 give a reward 0 when selected
- Arm 10 gives a reward 10
- You have 100 attempts
 - Which arms would you pull in attempts 1, 2, 3, ..., 100?

The k -Armed Bandit

- Arms 1 - 9 give a reward 0 when selected
- Arm 10 gives a reward 10
- You have 100 attempts
 - Which arms would you pull in attempts 1, 2, 3, ..., 100?
- What if you don't know anything about the arms?

Modeling An Arm

- Rewards are random variables $R^{(i)}, i = 1, 2, \dots, k$
- When you select arm i , it returns a reward (a number) that is drawn from a certain distribution
- Example distributions?

Stationarity

- Distribution corresponding to an arm isn't a function of time (attempt number)

Expected Reward on Selecting An Arm

- The unknown truth! $\phi^*(a), a=1,2,\dots,k$
- A conditional expectation, given the arm that was selected

$$\phi^*(a) = E[R_t | A_t = a]$$

- Time invariant --- Why?

arm

R_t is the reward you get at attempt t .
 A_t is the action, that is the arm chosen at attempt t .

Expected Reward on Selecting An Arm

- You don't know the truth
- Maintain and update estimates of the expected reward
 - Use the instantaneous rewards you get as you select arms

$Q_t(a)$ is your estimate for arm a
before you choose your t^{th} arm.

- Exploitation vs Exploration

TIME SEQUENCE OF ESTIMATES, REWARDS & ACTIONS

Convention:

R_t updates the estimate
 $Q_{t+1}(\cdot)$

At time t :

You have $Q_t(a) \forall a$.

Choose action A_t from $\{1, 2, \dots, K\}$

Receive reward R_t

Update Q_{t+1} for the action A_t was picked at t .

Consider $Q_t(a)$, given estimator is the sample mean

$$Q_t(a) = \frac{\sum_{j=1}^{t-1} I_{\{A_j=a\}} R_j}{\sum_{j=1}^{t-1} I_{\{A_j=a\}}}, \quad t=2, 3, \dots$$

Example:

Attempt No:	1	2	3	4	5
Action No:	4	1	4	2	4
Reward:	-1	0	5	10	-15

$Q_1(1) = Q_1(2) = Q_1(3) = Q_1(4) = 0$ [Initial values]

$$\begin{array}{l} Q_2(4) = \\ Q_2(1) = 0 \\ Q_2(2) = 0 \\ Q_2(3) = 0 \end{array} \quad \left| \quad \begin{array}{l} Q_3(4) = \cdot \\ Q_3(1) = \\ Q_3(2) = \\ Q_3(3) = \end{array} \right| \quad \begin{array}{l} Q_4(4) = \\ Q_4(1) \\ Q_4(2) \\ Q_4(3) \end{array}$$

An Example Estimate

- Sample Mean $\rightarrow$ generalization.

Show that:

$$Q^{(n)}(a) = \frac{(n-1)}{n} Q^{(n-1)}(a) + \frac{R^{(n)}(a)}{n}$$

n is the no. of times you picked a .

Generalization: $Q^{(n)}(a) = (1 - \alpha_n) Q^{(n-1)}(a) + \alpha_n R^{(n)}(a)$

